

Project Clean

AI-Based Content Moderation and Protection System for Social Platforms

Problem Statement !

In today's digital age, social media has become a popular space for communication and content sharing. However, this open environment has also led to a rise in harmful and inappropriate content, such as:

- Abusive language
- Political propaganda
- Deepfake videos
- Sexually explicit posts
- Targeted harassment of celebrities and VVIPs

A growing concern is the increasing number of teenage users, especially girls, sharing revealing or sensitive content to gain attention and followers, often unaware of the risks.

Current moderation systems lack transparency, struggle with multilingual content, and fail to provide age-appropriate control. Users who frequently violate rules often bypass filters, making platforms unsafe—especially for minors and vulnerable users.

Objective 🎯

Project Clean aims to develop a real-time, AI-powered content moderation system to ensure safe and responsible content sharing by:

- 🚫 Detecting and blocking sensitive content such as nudity, hate speech, political propaganda, and deepfakes.
- 👧 Enforcing age-based content controls, particularly protecting teenage girls from posting or accessing inappropriate material.
- 🗣️ Providing AI explainability to clarify why content was flagged or blocked.
- 🌐 Supporting multilingual abuse detection in regional Indian languages.
- 📉 Introducing a user reputation score to reduce repeated violations.
- 💬 Moderating comments and replies in real-time to filter out toxic behavior.

Expected Outcome

The system will provide the following key features:

1. **Automatic detection and blocking** of harmful content in text, images, and videos.
2. **Age-based content filtering**, especially protecting underage users.
3. **Support for Indian regional languages** in detecting abusive or harmful language.
4. **User reputation scoring**, discouraging frequent violators.
5. **Real-time comment moderation**, filtering out toxic replies.

6. **Explainable AI decisions**, showing why content was blocked.
7. **Admin dashboard** to manually review flagged content and manage users.

Technologies Used 🛠️

🚀 Frontend (User & Admin Panel)

- **React.js** – For building an interactive user interface
- **HTML, CSS, JavaScript** – For layout, styling, and client-side logic

🌐 Backend

- **Node.js + Express.js** – REST APIs and server-side logic
- **Python (Flask)** – AI/ML microservice for content analysis

🧠 AI/ML & Content Moderation

NLP (Text Analysis):

- Hugging Face Transformers (BERT, RoBERTa, MuRIL)
- spaCy – For preprocessing and entity detection
- Google Perspective API – Toxic comment analysis

Image/Video Analysis:

- OpenCV – Nudity and face detection
- NSFW.js / DeepFace / Deepfake detection models

🌐 Multi-language Support

- **IndicBERT / MuRIL** – Indian language understanding and abuse detection

🗄️ Database

- **MongoDB** – To store users, content, scores, and moderation logs

🧰 Other Tools

- **JWT / OAuth** – Secure authentication
- **Multer / Cloudinary** – File and image upload handling
- **Socket.IO (optional)** – Real-time comment moderation